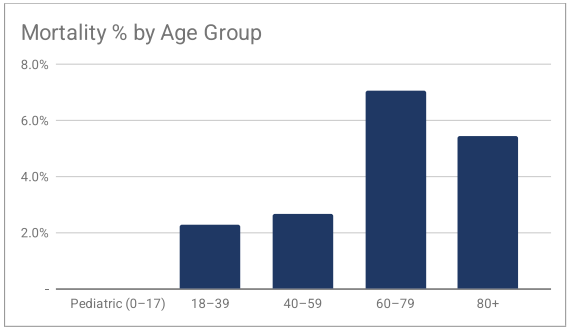
Inpatient Outcomes Analysis – Rheumatology 2018

All figures are live formulas on SLMCQC_IN2018 (one row per admission). Text matching is case-insensitive, so "DISCHARGED" / "DIscharged" are counted together.

1. OVERALL	Value
Total admissions	490
Unique patients	412
Discharged (incl. DAMA)	462
Expired	21
Other / blank (transferred, THOC, sti	7
Mortality rate	**4.3%**
Avg age – expired	62.4
Avg age – discharged	53.7
Admissions with readmission flag (#	24
Readmission rate	4.9%

KEY FINDINGS (update text if data changes)

Overall mortality 4.3% (21 of 490 admissions).
Patients who expired were older: avg 62.4 vs 53.7 yrs.
Highest-risk age group: 60–79 at 7.1%.
Male 5.2% vs Female 3.7% mortality.
Non-rheuma cases 12.5% vs rheuma 3.9% mortality.
185 admissions (38%) have no ICD10 code – incl. 14 of the deaths.



POSSIBLE ERROR SOURCES / DATA QUALITY (read before interpreting results)

185 admissions (38%) have no ICD10 code, including 14 of 21 deaths – diagnosis-level mortality is largely incomplete.
Only 21 deaths in total – rates for small groups (e.g. 1 death in 9) are unstable and may change a lot with one case.
2 admission(s) have an invalid Referral/Admission entry (e.g. a date) and are excluded from the encounter-type table.
2 admission(s) have a blank Rheuma/Non-rheuma field and are excluded from that table.
1 admission(s) have age stored as text and are excluded from age groups and average ages (the first record is read as a header by the QUERY formula in SLMCQC_IN2018!A1).
7 admission(s) have an outcome other than Discharged/Expired (transferred, THOC, still admitted or blank) – counted in totals but not as deaths.
Free-text inconsistencies (e.g. 'DISCHARGED' vs 'DIscharged', 'R ' with a space, 'A' vs 'Admission') are handled by case-insensitive / wildcard matching, but new spellings may be missed.
Admissions with several ICD10 codes count under each matching diagnosis, and readmissions rely on the manually entered # of Admissions column.

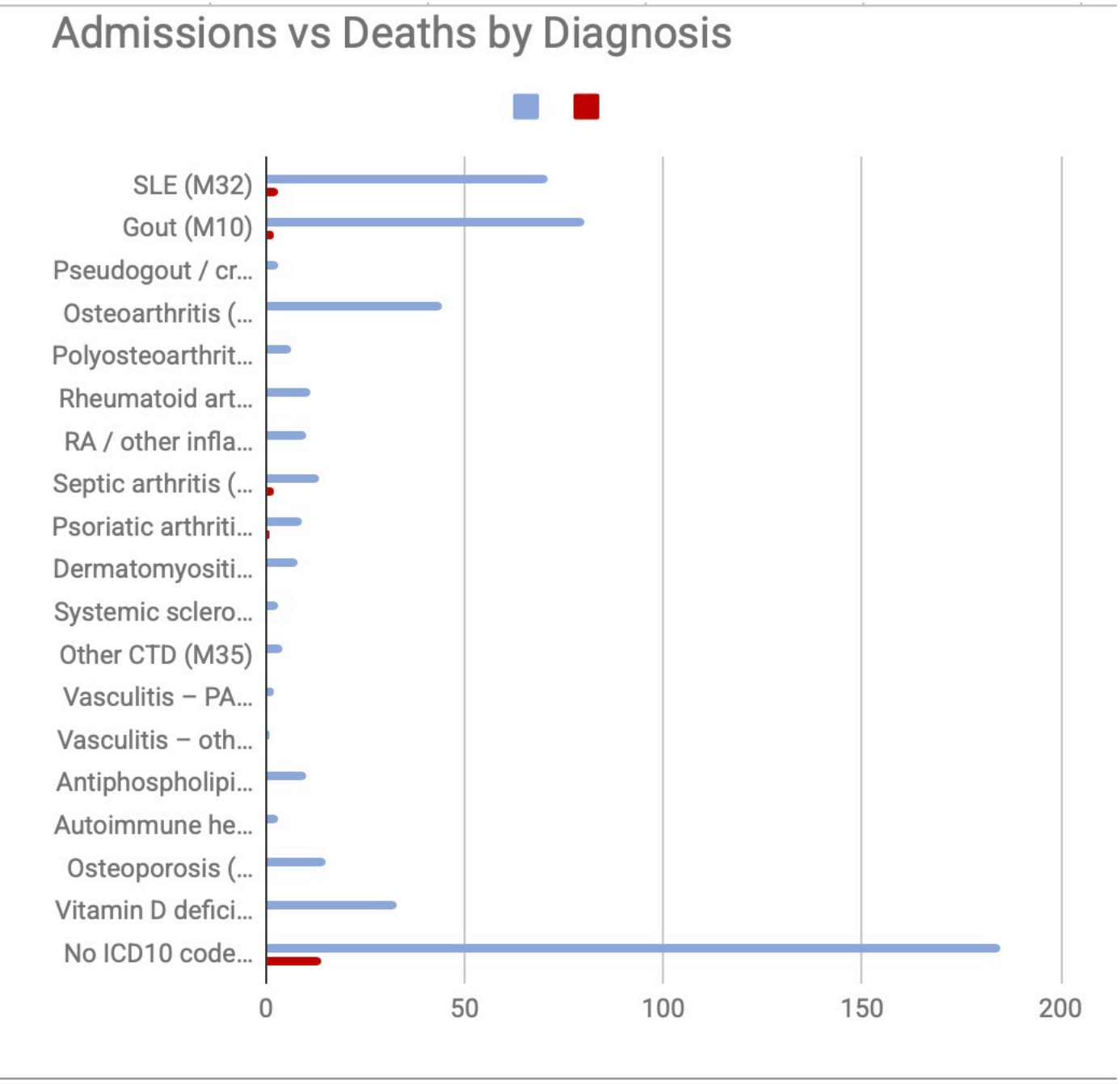
2. BY SEX	Match key	Admissions	Expired	Mortality %	Avg age	Readmit flag	Readmit %
Male	M	194	10	5.2%	56.7	6	3.1%
Female	F	296	11	3.7%	52.3	18	6.1%

3. BY AGE GROUP	Min age	Max age	Admissions	Expired	Mortality %	Avg age	Readmit flag	Readmit %
Pediatric (0–17)	0	17	18	0	-	9.7	0	-
18–39	18	39	133	3	2.3%	31.4	8	6.0%
40–59	40	59	113	3	2.7%	49.8	8	7.1%
60–79	60	79	170	12	7.1%	68.6	8	4.7%
80+	80	120	55	3	5.5%	87.3	0	-

4. BY ENCOUNTER TYPE	Match key	Admissions	Expired	Mortality %	Avg age	Readmit flag	Readmit %
Admission (A / Admission)	A*	143	6	4.2%	51.2	10	7.0%
Referral (R / Referral)	R*	345	15	4.3%	55.1	14	4.1%

5. RHEUMA vs NON-RHEUMA	Match key	Admissions	Expired	Mortality %	Avg age	Readmit flag	Readmit %
Rheuma case	R*	464	18	3.9%	53.5	24	5.2%
Non-rheuma case	NR	24	3	12.5%	65.3	0	-

6. BY DIAGNOSIS (ICD10)	Match key	Admissions	Expired	Mortality %	Avg age	Readmit flag	Readmit %
SLE (M32)	*M32*	71	3	4.2%	37.9	6	8.5%
Gout (M10)	*M10*	80	2	2.5%	61.1	3	3.8%
Pseudogout / crystal (M11)	*M11*	3	0	-	57.3	0	-
Osteoarthritis (M19)	*M19*	44	0	-	74.3	0	-
Polyosteoarthritis (M15)	*M15*	6	0	-	69.0	0	-
Rheumatoid arthritis (M05)	*M05*	11	0	-	57.8	0	-
RA / other inflammatory (M06)	*M06*	10	0	-	42.1	2	20.0%
Septic arthritis (M00)	*M00*	13	2	15.4%	61.5	0	-
Psoriatic arthritis (L40)	*L40*	9	1	11.1%	49.9	0	-
Dermatomyositis (M33)	*M33*	8	0	-	55.3	2	25.0%
Systemic sclerosis (M34)	*M34*	3	0	-	48.0	0	-
Other CTD (M35)	*M35*	4	0	-	65.0	0	-
Vasculitis – PAN (M30)	*M30*	2	0	-	78.0	0	-
Vasculitis – other (M31)	*M31*	1	0	-	61.0	0	-
Antiphospholipid syndrome (D68)	*D68*	10	0	-	31.5	0	-


Admissions vs Deaths by Diagnosis
SLE (M32)
Gout (M10)
Pseudogout / cr...
Osteoarthritis (...
Polyosteoarthrit...
Rheumatoid art...
RA / other infla...
Septic arthritis (...
Psoriatic arthriti...
Dermatomyositi...
Systemic sclero...
Other CTD (M35)
Vasculitis – PA...
Vasculitis – oth...
Antiphospholipi...
Autoimmune he...
Osteoporosis (...
Vitamin D defici...
No ICD10 code...
0
50
100
150
200

How to Do the Outcomes Analysis Manually in Python

Step-by-step guide to reproduce the Outcomes Analysis tab outside Google Sheets. Grey rows are code – copy them into Python exactly (keep the indentation).

⚠ Privacy first

This file has patient names, birthdays and hospital numbers. Remove the identifying columns before uploading the data anywhere (e.g. Google Colab, email). Step 4 shows how.

Step 1 – Export the data as CSV

1. Open the SLMCQC_IN2018 tab (the master list, one row per admission).
2. Go to File > Download > Comma-separated values (.csv). This downloads ONLY the tab you're on.
3. Rename the file to inpatient_2018.csv and save it in a folder you'll work from.

Note: this tab has NO header row – row 1 is already a patient. We'll add column names in Python (Step 3).

Step 2 – Set up Python

Easiest option: Google Colab (colab.research.google.com) – free, nothing to install. Upload the CSV with the folder icon on the left.
Or on your computer: install Anaconda (includes Python, Jupyter and pandas), then open Jupyter Notebook. If something is missing, run:

```
pip install pandas scipy matplotlib statsmodels
```

Step 3 – Load the CSV with column names

```
import pandas as pd
cols = ['PIN','Last','First','Middle','Birthday','Age','Sex','Phone','Date','Type','MD',
        'AdmDx','FinalDx','ICD10','Arthro','Joint','Steroid','HA','Polarizing','Ultrasound',
        'Immuno','Rheuma','Outcome','NumAdm']
df = pd.read_csv('inpatient_2018.csv', header=None, names=cols)
print(df.shape)      # should be about (490, 24)
df.head()
```

Step 4 – De-identify and clean

Drop names/phone, standardise text (e.g. 'DISCHARGED' vs 'DIscharged'), and create simple yes/no outcome columns.

```
df = df.drop(columns=['Last','First','Middle','Phone','Birthday'])
df['Outcome'] = df['Outcome'].astype(str).str.strip().str.title()
df['Died'] = (df['Outcome'] == 'Expired').astype(int)          # 1 = expired, 0 = not
df['Readmit'] = (pd.to_numeric(df['NumAdm'], errors='coerce') > 1).astype(int)
df['Age'] = pd.to_numeric(df['Age'], errors='coerce')
df['Sex'] = df['Sex'].astype(str).str.upper().str.strip()
df['Type'] = df['Type'].astype(str).str.upper().str[0].map({'A':'Admission','R':'Referral'})
df['ICD3'] = df['ICD10'].astype(str).str.strip().str[:3].replace({'nan':'Uncoded','':'Uncoded'})
print(df['Outcome'].value_counts())
```

Step 5 – Overall outcomes

```
print('Admissions:', len(df), '| Unique patients:', df['PIN'].nunique())
print('Deaths:', df['Died'].sum(), '| Mortality rate: {:.1%}'.format(df['Died'].mean()))
print(df.groupby('Died')['Age'].mean())     # avg age of survivors (0) vs expired (1)
```

Step 6 – Breakdowns (same as the tables on Outcomes Analysis)

One reusable function gives admissions, deaths, mortality %, average age and readmission % for any grouping.

```
def summary(by):
    t = df.groupby(by).agg(Admissions=('PIN','size'), Expired=('Died','sum'),
                           Mortality=('Died','mean'), AvgAge=('Age','mean'), ReadmitRate=('Readmit','mean'))
    return t.sort_values('Admissions', ascending=False).round(3)
summary('Sex')
summary('Type')
summary('Rheuma')
summary('ICD3').head(15)
```

Age groups – use pd.cut to make bins:

```
df['AgeGroup'] = pd.cut(df['Age'], bins=[0,17,39,59,79,120], labels=['0-17','18-39','40-59','60-79','80+'], include_lowest=True)
summary('AgeGroup')
```

Step 7 – Test whether relationships are real (statistics)

Chi-square test: is mortality different between groups (e.g. sex, rheuma vs non-rheuma)? p < 0.05 suggests a real difference.

```
from scipy import stats
table = pd.crosstab(df['Sex'], df['Died'])
chi2, p, dof, expected = stats.chi2_contingency(table)
print(table); print('p-value =', round(p, 4))
```

With few deaths (21), use Fisher's exact test for 2x2 tables instead – it's more reliable for small counts:

```
print(stats.fisher_exact(pd.crosstab(df['Rheuma'], df['Died'])))
```

T-test: is age different between patients who died and those who didn't?

```
print(stats.ttest_ind(df.loc[df.Died==1,'Age'].dropna(), df.loc[df.Died==0,'Age'].dropna(), equal_var=False))
```

Logistic regression: effect of several factors at once (odds ratio > 1 = higher risk of death).

```
import statsmodels.formula.api as smf, numpy as np
m = smf.logit('Died ~ Age + C(Sex) + C(Type)', data=df.dropna(subset=['Age','Type'])).fit()
print(m.summary()); print(np.exp(m.params))      # odds ratios
```

Caution: with only ~21 deaths, keep the model to 2-3 predictors, and treat results as exploratory.

Step 8 – Charts

```
import matplotlib.pyplot as plt
summary('AgeGroup')['Mortality'].plot(kind='bar', title='Mortality by Age Group'); plt.ylabel('Mortality rate'); plt.show()
df.boxplot(column='Age', by='Died'); plt.title('Age: survived (0) vs expired (1)'); plt.show()
```

Step 9 – Save results back to a file / Sheets

```
summary('ICD3').to_csv('mortality_by_diagnosis.csv')
```

Then in Google Sheets: File > Import > Upload > choose the CSV > 'Insert new sheet(s)'.

Common problems

- FileNotFoundError – the CSV isn't in the same folder as your notebook (or not uploaded to Colab). Check the file name exactly.
- Numbers look wrong / NaN – check text columns were cleaned (Step 4). Some ages or dates are typed as text.
- 185 rows have no ICD10 code – they appear as 'Uncoded'. Fill these in the sheet for a more accurate diagnosis analysis.
- Results differ slightly from the Outcomes Analysis tab – usually due to cleaning choices (e.g. rows with a date in the Referral/Admission column become blank).